DGMS 2019 GEOLOGY • QUESTION 33

Subduction Tectonics: Underplating vs. Offscraping in Accretionary Prisms

Topic: Structural Geology & Global Tectonics | Primary Reference: Van der Pluijm & Marshak (Earth Structure, Ch. 17)

Question Statement

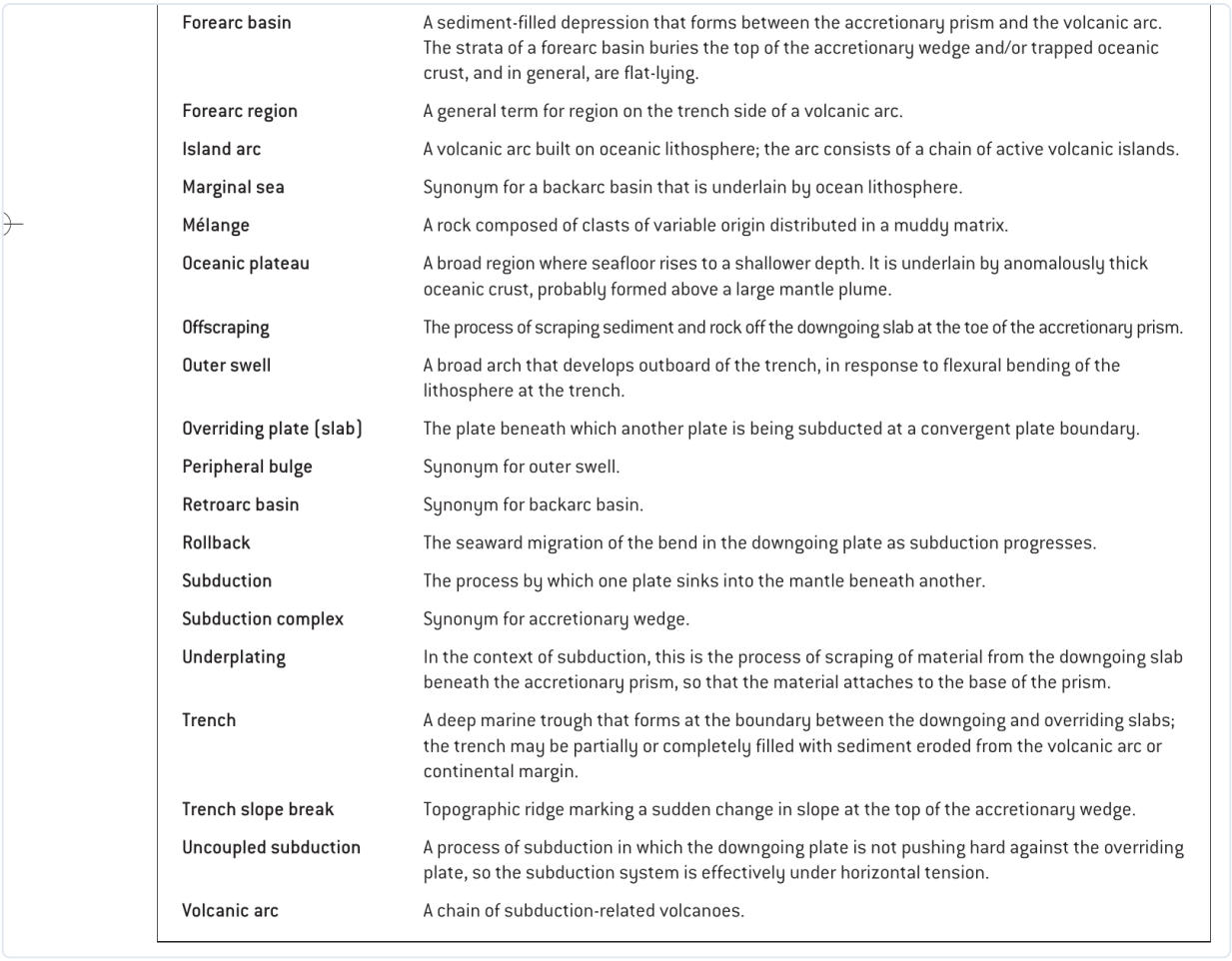
Q.33. The rocks and sediments that were subducting sometimes get attached with overriding plates by thrusting. This process is commonly called

(a) Obduction
(b) Rejuvenation
(c) Underplating
(d) Offscraping

a Obduction	b Rejuvenation
c **Underplating (Correct Answer)**	d Offscraping

✓ Correct Answer: (c) Underplating [Note: In subduction tectonics, underplating specifically denotes the basal accretion of subducting material to the underside of the overriding plate by thrusting; offscraping denotes frontal accretion at the trench toe].

Textbook Evidence: Van der Pluijm & Marshak (Chapter 17)



Forearc basin	A sediment-filled depression that forms between the accretionary prism and the volcanic arc. The strata of a forearc basin buries the top of the accretionary wedge and/or trapped oceanic crust, and in general, are flat-lying.
Forearc region	A general term for region on the trench side of a volcanic arc.
Island arc	A volcanic arc built on oceanic lithosphere; the arc consists of a chain of active volcanic islands.
Marginal sea	Synonym for a backarc basin that is underlain by ocean lithosphere.
Mélange	A rock composed of clasts of variable origin distributed in a muddy matrix.
Oceanic plateau	A broad region where seafloor rises to a shallower depth. It is underlain by anomalously thick oceanic crust, probably formed above a large mantle plume.
Offscraping	The process of scraping sediment and rock off the downgoing slab at the toe of the accretionary prism.
Outer swell	A broad arch that develops outboard of the trench, in response to flexural bending of the lithosphere at the trench.
Overriding plate (slab)	The plate beneath which another plate is being subducted at a convergent plate boundary.
Peripheral bulge	Synonym for outer swell.
Retroarc basin	Synonym for backarc basin.
Rollback	The seaward migration of the bend in the downgoing plate as subduction progresses.
Subduction	The process by which one plate sinks into the mantle beneath another.
Subduction complex	Synonym for accretionary wedge.
Underplating	In the context of subduction, this is the process of scraping of material from the downgoing slab beneath the accretionary prism, so that the material attaches to the base of the prism.
Trench	A deep marine trough that forms at the boundary between the downgoing and overriding slabs; the trench may be partially or completely filled with sediment eroded from the volcanic arc or continental margin.
Trench slope break	Topographic ridge marking a sudden change in slope at the top of the accretionary wedge.
Uncoupled subduction	A process of subduction in which the downgoing plate is not pushing hard against the overriding plate, so the subduction system is effectively under horizontal tension.
Volcanic arc	A chain of subduction-related volcanoes.

Figure 17.4: Anatomy of a Convergent Margin Accretionary Prism (Van der Pluijm & Marshak, *Earth Structure: An Introduction to Structural Geology and Tectonics*, 2nd ed., p. 433). Cross-section illustrating the mechanics of subduction accretion: incoming trench sediments are either scraped off frontally at the trench toe (**offscraping**) or carried beneath the prism along the subduction channel and transferred to the base of the overriding plate by basal thrust faults and duplex slicing (**underplating** / **basal accretion**).

> *"Sediment can be added to an accretionary wedge in two ways: by* **offscraping** *at the toe of the wedge, or by* **underplating** *(also called basal accretion) beneath the wedge. In underplating, sediment and slivers of oceanic crust that were subducted to depths of several kilometers or more are transferred to the overriding plate along thrust faults that branch off the subduction décollement."*
>
> **— Ben A. van der Pluijm & Stephen Marshak, *Earth Structure*, Chapter 17 (Convergent Margins), p. 433**

Comprehensive Option Analysis

(c) Underplating (Subduction Underplating / Basal Accretion) **[CORRECT]**

Mechanics & Tectonic Definition: When oceanic lithosphere carrying a veneer of sediment subducts beneath an overriding plate, not all material is scraped off immediately at the trench toe. A substantial volume of sediments and fragments of oceanic crust travels down the *subduction channel*. As pressure, dewatering, and frictional resistance increase at depths between 5 km and 30 km, slices of these **subducting rocks and sediments** are detached from the downgoing slab by thrust faults that step down into the footwall. These slices are welded and accreted to the base (underside) of the overriding wedge via duplex structures. This basal accretionary attachment of previously subducting material is precisely defined as **underplating** (or subduction underplating).

Diagnostic Clue in Question: The phrasing *"The rocks and sediments that were subducting sometimes get attached with overriding plates by thrusting"* explicitly captures rocks that were already underway in the subduction channel being transferred at depth, which is the textbook distinction between underplating (basal accretion) and offscraping (frontal accretion at the toe).

(d) Offscraping (Frontal Accretion) **[INCORRECT / DISTINCT PHENOMENON]**

Definition: *Offscraping* refers specifically to **frontal accretion** at the toe of the trench. Before the sediment can subduct beneath the overriding plate, the upper layers of trench turbidites and pelagic muds are peeled off by seaward-verging or landward-dipping imbricate thrusts right at the deformation front.

While both offscraping and underplating are subduction accretion mechanisms, offscraping occurs at the *trench surface (toe)* before subduction to depth, whereas underplating captures material that *was subducting* down the interface and becomes attached to the base of the overriding plate.

(a) Obduction **[INCORRECT]**

Definition: *Obduction* is the tectonic emplacement of a large slab of dense oceanic lithosphere (an **ophiolite suite**) onto a buoyant continental margin or island arc during an arc-continent or continent-continent collision (e.g., the Semail Ophiolite in Oman). It involves catastrophic overthrusting of an entire oceanic lithospheric

section onto continent, rather than routine accretionary transfer of subducting sediments during steady-state subduction.

(b) Rejuvenation **[INCORRECT]**

Definition: In geology, *rejuvenation* refers to: (1) Geomorphic renewal of stream erosive power caused by base-level fall or tectonic uplift (forming entrenched meanders and river terraces), or (2) Rejuvenated volcanism (post-erosional volcanic stage on oceanic hotspot islands like Oahu, Hawaii). It has no relation to subduction thrust accretion.

Comparison: Accretionary Wedge Mass-Transfer Mechanisms

Process	Location / Depth	Structural Style	Material Involved
Offscraping (Frontal Accretion)	Trench toe / Deformation front (0–5 km depth)	Imbricate thrust fans, protothrusts, frontal folds	Unconsolidated trench turbidites, hemipelagic muds
Underplating (Basal Accretion)	Base of overriding plate (5–30 km depth)	Antiformal duplexes, basal shear zones, underthrust nappes	Subducted sediment, basaltic ocean crust, chert, melange
Subduction Erosion (Tectonic Erosion)	Trench toe or subduction channel	Basal scouring, extensional collapse of forearc	Forearc crust ground up and dragged down into the mantle
Obduction	Continental margin collision zone	Major crustal sutures, continental-scale thrust sheets	Complete ophiolite suite (harzburgite, gabbro, sheeted dikes, pillow basalt)

Key Takeaway & Exam Strategy

Competitive Exam Rule:

- When a question mentions transfer of *subducting* material to the *underside/base of the overriding plate by thrusting*, the technically rigorous answer is **Underplating** (basal accretion).

- When a question refers to *peeling or scraping of trench sediments at the front/toe of the trench*, the term is **Offscraping**.
- Both processes together constitute **Subduction Accretion**, which enlarges the forearc accretionary prism over geologic time.